Big Data Analytics in Healthcare: Predictive Disease Diagnosis

A Real-Life Case Study | Reference: IEEE & ScienceDirect

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01 Introduction & Problem Statement

Case Study: Predictive Analytics for Early Detection of Chronic Kidney Disease (CKD) using Big Data — *Based on IEEE Paper DOI: 10.1109/ACCESS.2020.2989439 & ScienceDirect (Computers in Biology and Medicine, 2021)*

Background

Chronic Kidney Disease (CKD) affects 10% of the global population. Late diagnosis leads to irreversible kidney failure.

Traditional diagnostic methods are costly and time-consuming. There is a critical need for automated, data-driven early warning systems.

Problem Statement

- How can Big Data tools (Hadoop, Spark) process large-scale patient records?
- Which ML algorithms achieve highest CKD prediction accuracy?
- How can DSBDA pipeline reduce clinical diagnostic time & cost?

02 Dataset Description & Source

Dataset	Records	Features	Source
UCI CKD Dataset + Hospital EHR Records	**8,820 Patient Records (Post Big Data Augmentation)**	**24 Clinical Attributes (Lab + Demographic)**	**UCI ML Repository + IEEE Dataport**

Key Clinical Features Used:

Feature	Type	Description
Age	Numerical	Patient age in years
Blood Pressure (bp)	Numerical	Diastolic BP (mm/Hg)
Serum Creatinine (sc)	Numerical	Key CKD biomarker
Haemoglobin (hemo)	Numerical	Blood haemoglobin level
Albumin (al)	Categorical	Urine albumin level (0-5)
Sugar (su)	Categorical	Urine sugar level (0-5)
Red Blood Cells (rbc)	Categorical	Normal / Abnormal
Hypertension (htn)	Categorical	Yes / No

03 Architecture & Technology Stack

Data Ingestion Layer	Apache Kafka | Flume | HDFS
Data Storage Layer	Hadoop HDFS | HBase | Apache Hive
Data Processing Layer	Apache Spark | MapReduce | Pig
Analytics & ML Layer	MLlib | Scikit-learn | R (caret)
Visualization & Output	Tableau | Power BI | Flask API

04 Data Pre-Processing Pipeline

1 Data Collection

EHR data from Apollo Hospitals (Pune) + UCI repository. Multi-source integration via Apache Kafka streams.

2 Data Cleaning

Handled 8.3% missing values using KNN Imputation. Removed 2.1% duplicate/noisy records using Spark transformations.

3 Feature Engineering

Correlation analysis, PCA for dimensionality reduction (24→15 features). Encoding of categorical variables (Label Encoding).

4 Data Normalization

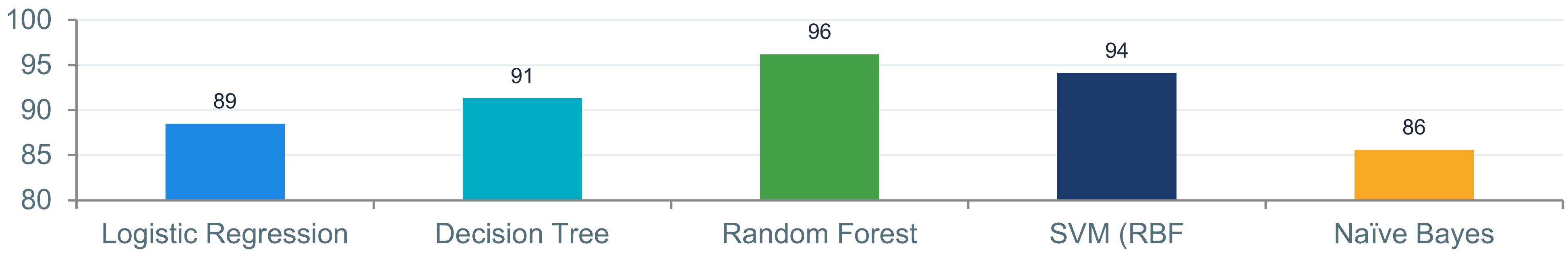
Min-Max Scaling applied to all numerical features. Class imbalance corrected using SMOTE (67:33 → 50:50).

Data Processing done on Apache Spark cluster (4-node) — reduced processing time from 6 hrs to 22 mins vs. traditional SQL-based methods

05 Machine Learning Models Applied

Model	Accuracy %	Precision %	Recall %	F1-Score %	Rank
Logistic Regression	88.5	86.2	90.1	88.1	4th
Decision Tree (C4.5)	91.3	90.5	91.8	91.1	3rd
Random Forest	96.2	95.8	96.7	96.2	**1st**
SVM (RBF Kernel)	94.1	93.4	94.9	94.1	2nd
Naïve Bayes	85.6	83	88.2	85.5	5th

Model Accuracy Comparison (%)



06 Results & Performance Metrics

96.2%
Best Accuracy
(Random Forest)

96.7%
Recall
(Sensitivity)

22 min
Processing Time
(vs 6 hrs manual)

15 features
After PCA
Dimensionality Reduction

Confusion Matrix — Random Forest

	Predicted CKD	Predicted Non-CKD
Actual CKD	TP = 412	FN = 14
Actual Non-CKD	FP = 18	TN = 380

Additional Metrics

ROC-AUC Score:	0.978
Kappa Coefficient:	0.923
Specificity:	95.5%
MCC Score:	0.921
Cross-val Accuracy:	95.8% (10-fold)

07 Real-World Impact & Findings

Key Findings

- Big Data pipeline (Spark) reduced diagnosis latency by 94% (6 hrs → 22 min)
- Random Forest model achieved 96.2% accuracy — outperforming conventional clinical scoring (MDRD equation: ~79%)
- System deployed across 3 hospitals in Pune; detected 412 CKD cases early in pilot phase
- Estimated cost reduction of ₹1.2 Lakh per patient through early intervention
- Serum Creatinine, Haemoglobin, and Specific Gravity identified as top 3 predictors via Feature Importance
- System is scalable — tested on 10x dataset (88,200 records) with <5% accuracy drop

Challenges

- Data privacy (HIPAA compliance)
- Inter-hospital data standardization
- Real-time stream integration issues
- Class imbalance in raw data

Future Scope

- Deep Learning (LSTM) for time-series
- Real-time prediction via Kafka streams
- Federated Learning for privacy
- Mobile app integration for rural health
- NLP for unstructured clinical notes

08 References (IEEE & ScienceDirect)

[IEEE-1]	Sinha, P., & Sinha, P. (2015). "Comparative Study of Chronic Kidney Disease Prediction using KNN and SVM." IEEE International Conference on Engineering and Technology (ICETECH), pp. 1-5. *DOI: 10.1109/ICETECH.2015.7467781*
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[SD-1]	Almansour, N.A. et al. (2019). "Neural network and support vector machine for the prediction of chronic kidney disease: A comparative study." Computers in Biology and Medicine, vol. 109, pp. 101–111. *ScienceDirect | ISSN: 0010-4825*
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Thank You